



River Trothy 2023-25

A report on behalf of the Wye Salmon Association.

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A review of citizen science data recorded in the Trothy sub catchment between Aug-23 and Dec-25.

Introduction

This report reviews citizen science water quality data obtained from the Trothy sub catchment between August 2023 and December 2025. The data has been recorded exclusively by Friends of the River Wye FORW volunteers and includes Hanna phosphate¹, specific conductivity (SC), temperature, total oxidised nitrogen (TON) and nitrite using test strips, and turbidity using a Secchi tube.

River Trothy sub catchment

Overview

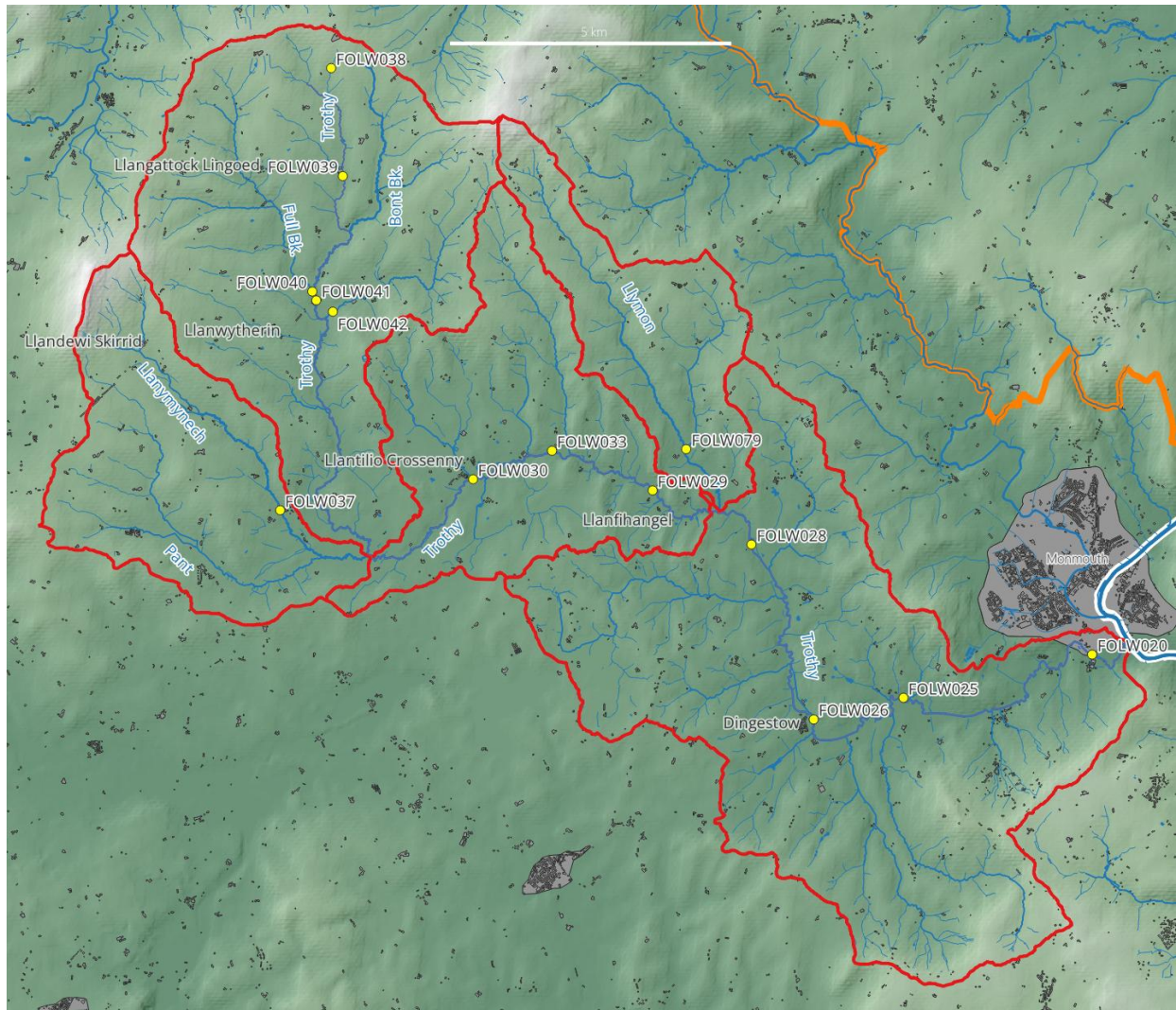


Figure 1: the Trothy sub catchment (red outline). Yellow circles indicate the sampling locations referred to in this study, with their WyeViz reference number.

The River Trothy drains 147 km² of Monmouthshire farmland, located in the area between Abergavenny and Monmouth. Total length of the main river (straightened out) is 37km. The Trothy catchment, shown in Figure 1, is for the most part gently rolling countryside. The two high spots visible in figure 1 (at the headwaters of the Llanymynach and Llymon brooks) both exceed 400m elevation. But for the most part, the sub catchment lies below 150m. Other than the outskirts of Monmouth, there are no large settlements falling within the sub catchment.

¹ Throughout this report, we shall report reactive phosphorus (RP) as mgP/l, rather than the raw Hanna reading in mgPO₄/l.

Hydrology

Until 1982, the Trothy had a gauging station at Mitchel Troy. Data from this is still available on the [CEH National River Flow Archive](#), providing some key statistics. Annual rainfall on the sub catchment was about 960mm, and annual average flowrate at Mitchel Troy was 1.56 m³/s. The river is moderately “flashy”, having a base flow index (BFI) of 0.49 (compared to 0.73 for the Garren a few km northeast). Median flow (Q_{50}) is 0.63 m³/s.

Figure 2 shows the elevation profile for the Trothy. The headwater feeder streams fall off land reaching to above 300m elevation, but the river proper commences from about 160m. The first 5km see the river losing elevation at around 16m per km. But from the 10km mark to the Wye confluence, the gradient is fairly consistent at around 2m per km. The higher elevation headwaters of the upper catchment (Llanymynach and Llymon, as well as Trothy) are undoubtedly driving the relatively low BFI.

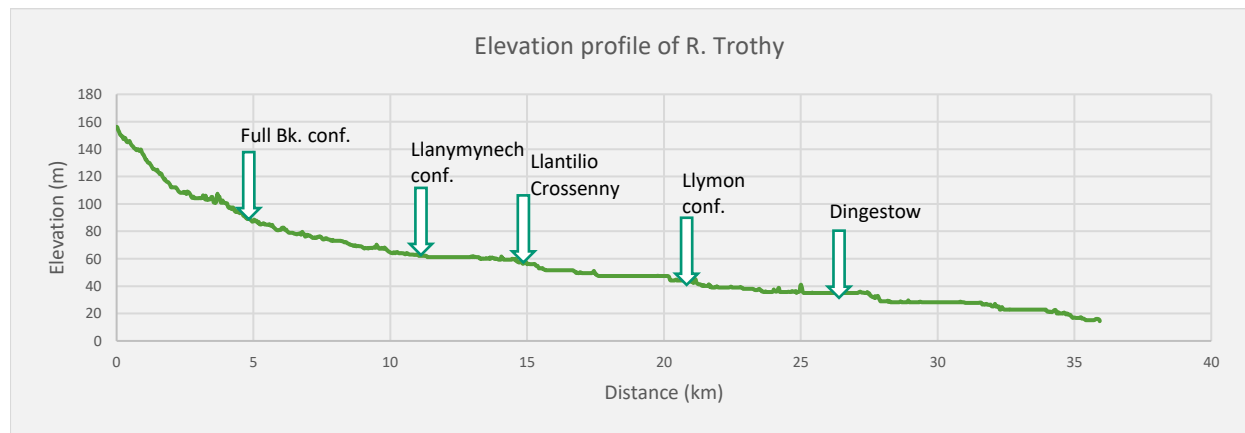


Figure 2: elevation profile for the main stem of the Trothy.

For some of our analysis later, we shall require daily flow data. In the absence of current data for the Trothy, we must use proxy data and, given its geographical proximity, the Marstow Mill gauging station on the Garren Brook seems appropriate. Although the BFI is higher on the Trothy, this would primarily seem to relate to peak flows from the high elevation headwaters. The general topography of most of the Trothy catchment is, in fact, reasonably similar to the Garren catchment, also rainfall is similar, meaning that one could expect broadly proportionate low to medium (groundwater driven) flow behaviours. The median flows are, in fact, almost identical between the two rivers and Q_{95} values are also similar. Thus, in the absence of other options, we shall use Marstow Mill as a proxy for Trothy flows in the analysis to follow. Further details on this topic are to be found in Appendix A.

Land use

Land use within the Trothy sub catchment is shown in figure 3. This shows a strong bias towards livestock farming, with 76% of land classified as improved grassland and only 13% of land being used for arable farming. A little under 10% of land is occupied by woodland and there is a tiny amount of acid grassland above 350m elevation on the two hills identified above. This represents a significant shift from 2000 when only 59% was improved grassland and 15% remained unimproved acid grassland. This agricultural intensification implies an increase in nutrient loads within the sub catchment.

Essentially, the catchment is full of cattle and sheep. Pro-rata to Welsh national statistics, the area of improved grassland would suggest perhaps 4,500 cattle and 100,000 sheep, which is a lot of manure. But we are aware of no large intensive poultry operations within the sub catchment.

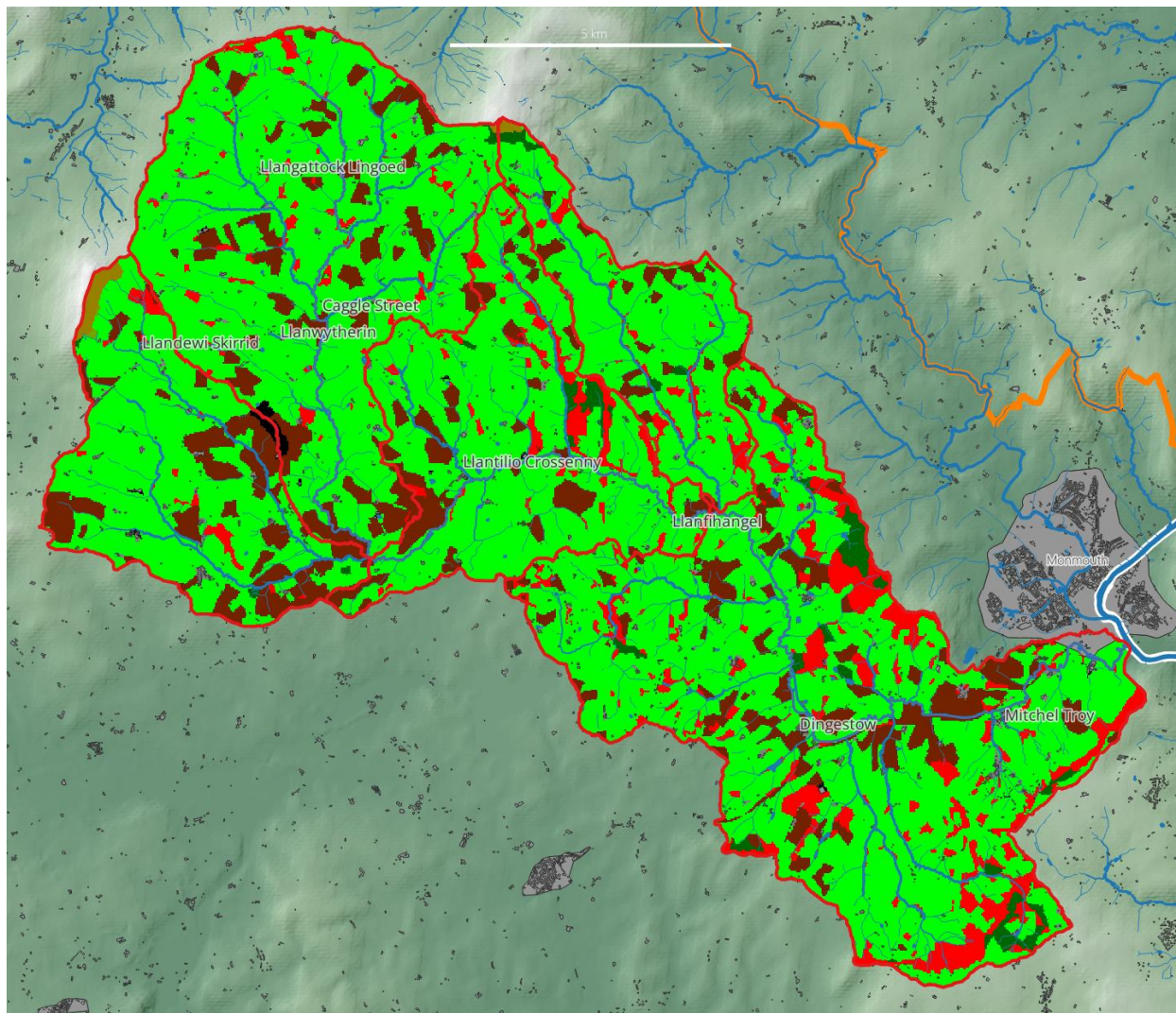


Figure 3: land use in the Trothy sub catchment using 2023 GIS data from CEH. Green is improved grassland (76%), brown is arable/horticulture (13%), red is broadleaved woodland (8%), dark green is coniferous woodland (1.4%) & olive brown is acid grassland (0.3%).

Statutory targets

The Trothy is not part of the Wye SAC, therefore Water Framework Directive (WFD) targets apply. Using the formula specified in *The Water Framework Directive (Standards and Classification) Directions (England and Wales) 2015*, we can calculate the WFD requirements for RP in different parts of the sub catchment as follows.

Location	Maximum RP (mgP/l)		
	High	Good	Moderate
Llymon at Trothy confluence	0.045	0.083	0.201
Llanymynach at Trothy confluence	0.043	0.079	0.194
Upper Trothy (above Llanymynach)	0.040	0.074	0.183
Lower Trothy (Llymon to Wye confluence)	0.048	0.087	0.208

Table 1: Water Framework Directive classification limits at different points in the Trothy sub catchment. Values are calculated as a function of site elevation and river alkalinity.

Since the national target is to achieve *good* status, WFD essentially requires less than around 0.09 mgP/l in the lower river and less than around 0.08 mgP/l upstream (as annual and growing season means, not for each measurement).

WFD does not specify a target for nitrate.

Sampling locations

FORW sampling locations, as shown in figure 1, are summarised in Table 2. Sites with limited sampling, specifically FOLW021, 033, 038, 039 & 042, have been excluded from this analysis. At many sites, regular sampling commenced part way through 2023, explaining the lower sample counts for that year. Consistent, mostly weekly sampling continued through 2024-5 at sites on the lower Trothy, but on the middle Trothy sampling fell off in 2025. Sites on the upper Trothy, Llymon and Llanymynach have been more sporadic, however those listed can potentially still give us useful annual averages, if only for one year each.

		Number of samples by year		
Reference	Site name	2023	2024	2025
Llymon Brook:				
FOLW079	Llymon Brook	-	-	28
Llanymynach Brook:				
FOLW037	Trothy, Llanymynach Br. Llanfable Br.	20	-	-
Full Brook:				
FOLW040	Trothy / Full Brook at Pont Gilbert	5	18	4
Upper Trothy (above Llanymynach confluence):				
FOLW041	Trothy at Pont Gilbert	5	17	6
Middle Trothy (Llanymynach to Llymon confluences)				
FOLW030	Trothy at Llantilio Crossenny	19	42	7
FOLW029	Trothy at Onen Bridge	7	42	6
Lower Trothy (below Llymon confluence)				
FOLW028	Trothy at Abbey Bridge	21	47	51
FOLW026	Trothy at Dingestow	22	52	43
FOLW025	Trothy at Jingle Street	3	37	40
FOLW020	Trothy by Troy House	21	48	46

Table 2: summary of FORW sampling sites possessing reasonable datasets in the Trothy sub catchment.

Point sources

There are six STWs discharging into the streams of the Trothy sub catchment, as listed in table 3.

Name	Discharging to	Input PE
Llanddewi Rhydderch	River Trothy	100
Llanvapley	Pant Brook	19
Llantillio Crossenny	River Trothy	28
Great Craig	Llymon Br	42
Penrhos	Llanymynach Brook	15
Dingestow	River Trothy via Nant Wechan	232
Total		436

Table 3: STWs operated by DC-WW which discharge into the River Trothy or its tributaries. Data is taken from a 2024 FoI request by WSA. (PE, meaning population equivalent, is a standardised unit representing the average output from one person).

In addition, we need to include private treatment systems. The list in table 4 is taken from current NRW permitting data. But sadly, this is another area which our regulators choose not to regulate effectively. There is a huge legacy of old, non-compliant, often poorly maintained, domestic septic tanks which discharge into surface water illegally, and which are conveniently not known to regulators. Studies in the scientific literature have suggested that regulators' records represent as little as 5% of this category.

Taking tables 3 and 4 together suggests that licensed sewage effluent entering the Trothy and its tributaries totals approximately 750 PE.² The locations of these discharges are shown in figure 4. However, based on the average Welsh farm size of 125 ha, there are in the region of 100 farms in this catchment and it implausible that a good fraction of these is not discharging to surface water. Thus, a

² PE (population equivalent), is a standard capacity unit in water treatment, implying 180 l/day of effluent, and incorporating 1.4g/day of phosphorus.

rough estimate based on 50% of farms at 5 PE per farm would suggest the total effluent load is potentially in excess of 1000 PE.

Permit No.	Site name	Discharging to	Input PE
AN0072901	CROSS ASH PRIMARY SCHOOL	LLYMON BROOK	8
AN0106801	THE STEPPES NURSING HOME	TREREW BROOK	14
AN0222201	BRIDGE CARAVAN PARK DINGESTOW MONMO	RIVER TROTHY	82
AN0237801	THE ROLLS OF MONMOUTHSHIRE LTD	UNNAMED TRIB OF RIVER TROTHY	50
AN0261801	CROFT FARM LLANVERTHERINE NR ABERG	TRE-RHEW BROOK	7
AN0309001	HIGHWAYS DIRECT WORKS TROY DEPOT	RIVER TROTHY	6
AN0365801	GREAT LLAN THOMAS FARM CWMCARVAN	A TRIB OF THE RIVER TROTHY	12
AN0391201	FARMHOUSE AT BAILEY GLACE FARM	CWMCARTHEN BROOK	5
AN0397701	FARMHOUSE AT BAILEY GLACE FARM	TRIBUTARY OF CWMCARTHEN BROOK	7
CB3799FB	2 Royce Barns	UNNAMED TRIB OF RIVER TROTHY	10
WP3029XJ	UPPER PANT FARM	TRIBUTARY OF THE PANT BROOK	8
WP3122GW	CHERRY ORCHARD COTTAGE STP	TRIBUTARY OF FULL BROOK	6
ZP3026XR	STP SERVING CAER-LLAN	UNNAMED TRIB CWMCARFON BRK	103
Total			316

Table 4: NRW permitted private treatment plants discharging into the Trothy and its tributaries. The PE column is calculated from the maximum daily total flow specified in the permit, divided by 180 l/d per PE. Note that non-residential premises, including schools and businesses, have fractional PE ratings reflecting their part-time occupancy.

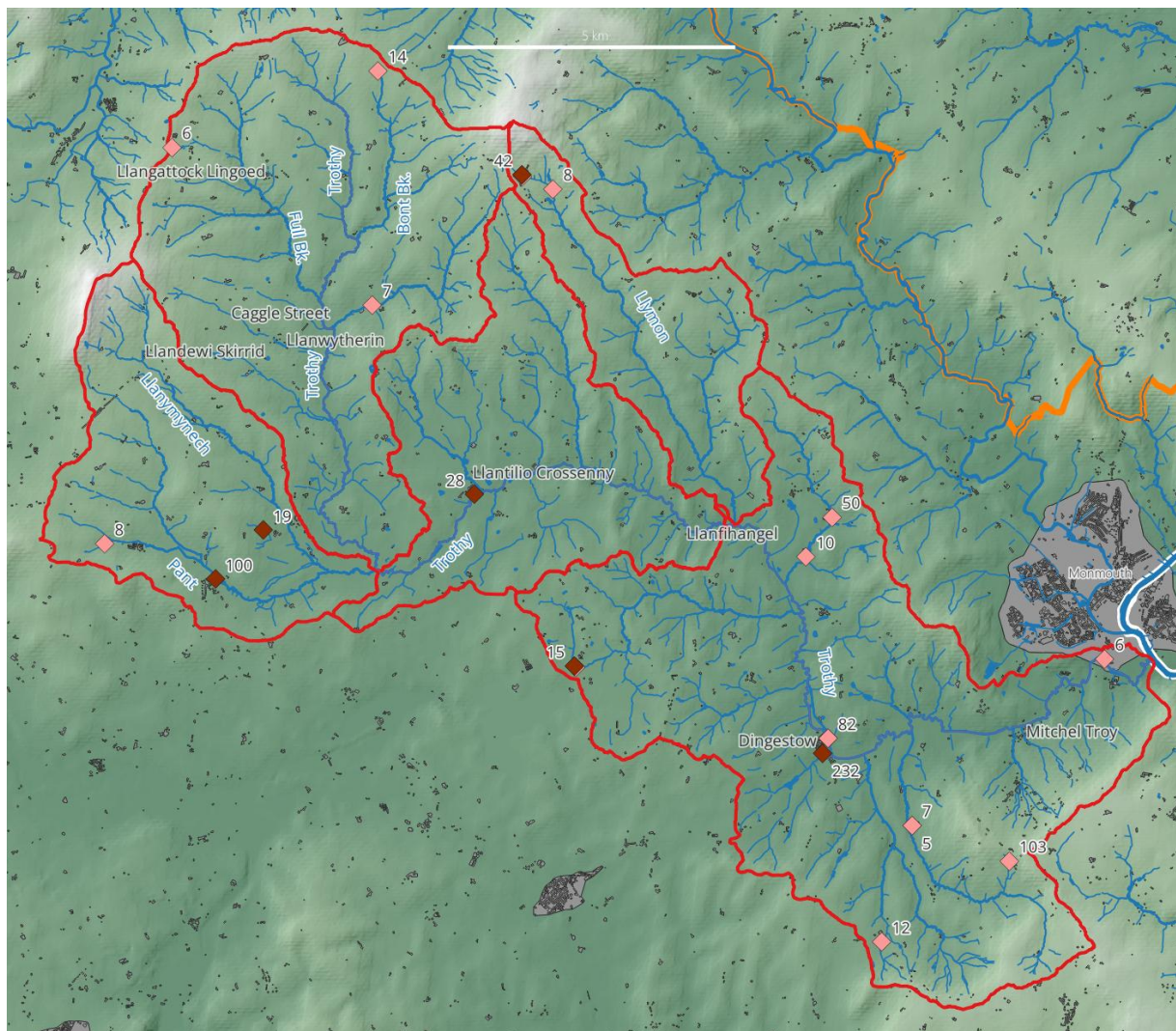


Figure 4: NRW-permitted sewage systems discharging to surface water within the Trothy sub catchment. Brown diamonds are DC-WW STWs; pink diamonds are private treatment systems; associated numbers are the stated input load in PE.

Results and discussion

Summary statistics

Data from the Trothy sub catchment over 2023-25 totals 713 samples. However, as explained above, some locations have been abandoned without accumulating a full year's data. The Llanymynach and several upper Trothy sites are in this category. The following analysis will therefore focus on the lower Trothy (below the Llymon), the middle Trothy (Llanymynach to Llymon) and the Llymon.

Mean values for the key measured parameters are shown in table 3.

Reference	Site name	Year	SC (μS/cm)	RP (mgP/l)	Turbidity (NTU)	TON (mgN/l)	Samples
Llymon Brook:							
FOLW079	Llymon Brook	2024					-
		2025	506	0.11	16	2.9	28
Middle Trothy (Llanymynach to Llymon confluences)							
FOLW030	Trothy at Llantilio Crossenny	2024	383	0.07	25	3.0	42
		2025					7
FOLW029	Trothy at Onen Bridge	2024	395	0.07	26	2.6	42
		2025					6
Lower Trothy (below Llymon confluence)							
FOLW028	Trothy at Abbey Bridge	2024	439	0.10	38	2.0	47
		2025	491	0.11	23	2.2	52
FOLW026	Trothy at Dingestow	2024	453	0.08	30	3.4	52
		2025	505	0.10	21	3.5	43
FOLW025	Trothy at Jingle Street	2024	447	0.12	46	1.4	37
		2025	498	0.16	30	2.8	40
FOLW020	Trothy by Troy House	2024	447	0.10	49	2.5	48
		2025	496	0.11	25	2.9	46

Table 3: annual mean values for SC, RP, turbidity & TON over the period Jan-24 to Dec-25. Only sites with a useful quantity of data are included here. Nitrite has not been included because it is generally recorded as zero.

Specific conductivity is consistent across all sites: around 450 μ S/cm in 2024, increasing to around 500 μ S/cm in 2025. Because this parameter is driven primarily by dissolved mineral ions, the increase in 2025 almost certainly reflects the lower summer flows seen that year; a point we shall explore further below.

The two middle Trothy sites showed marginal WFD *good* status for annual mean RP in 2024. However, this was not replicated at the lower Trothy sites which fall in the *moderate* range (compare to table 1). Moreover, these sites all became noticeable worse in 2025; the reason for which will be examined below. The Llymon annual mean RP value for 2025 was similar to the lower Trothy sites.

Annual mean turbidity is of questionable utility because the low flow and high flow regimes are very different. That said, these values are a good deal higher than those for the Worm Brook, which for 2025 are in the 5-15 NTU range. However, this could simply be a consequence of the use of Secchi tubes on the Trothy, which have a minimum reading of 12 NTU.

Individual TON readings are generally 2 or 5 mgN/l, leading to the annual mean values falling in the 2-3 mgN/l range. Although TON is not a WDF determinand, EA have suggested that 1.5 mgN/l would be reasonable target, which puts all these locations in the "too high" category. Nevertheless, the values for the Trothy are substantially below the 7-9 mgN/l averages seen on the Worm Brook in 2025; the latter sub catchment having a significant fraction of arable farming.

We shall now examine each parameter in more detail.

Specific conductivity

As an exemplar, figure 5 shows the 2025 trend plot from FOLW020, Troy House, at the very bottom of the sub catchment. This shows a slowly rising trend throughout the drier months, with erratic changes during the winter period. We can explore this further by plotting against Marstow Mill discharge as per figure 6.

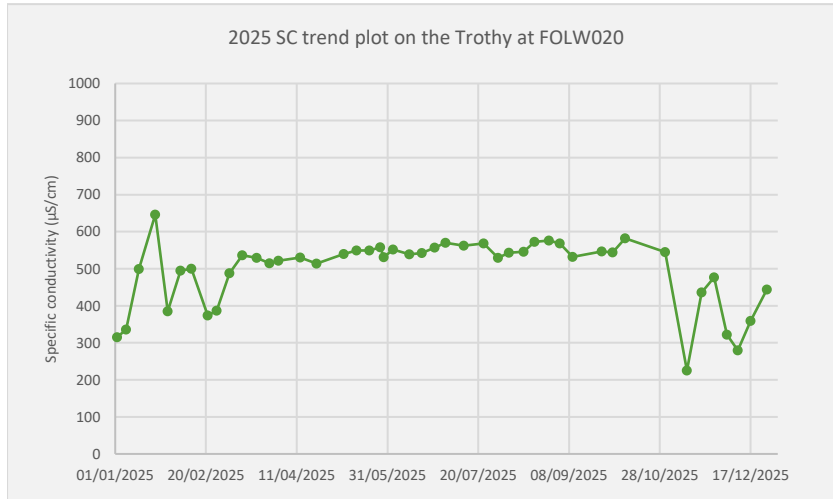


Figure 5: Specific conductivity trend plot for FOLW020, Troy House, during 2025.

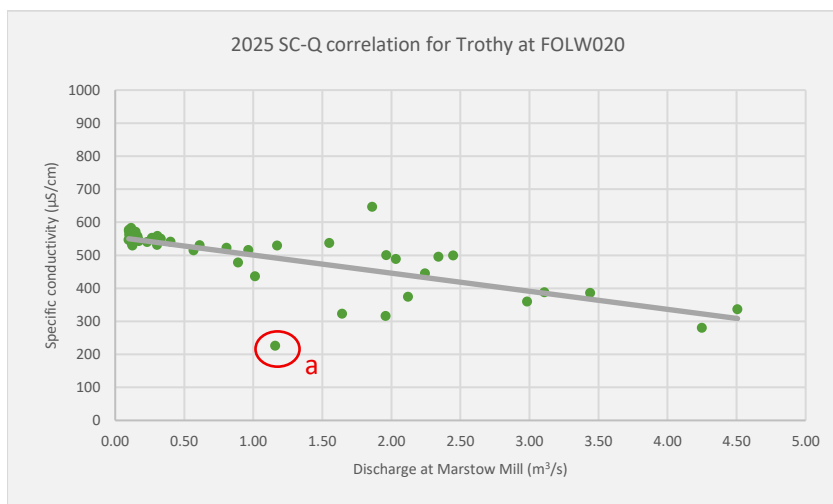


Figure 6: SC-Q correlation plot for FOLW020, Troy House, during 2025.

Figure 6 shows the usual, approximately linear fall-off of SC with discharge. This arises primarily because, during wet weather, groundwater passes more rapidly to the river with less opportunity to pick up mineral ions from soil and bedrock. The outlier circled at 'a' is interesting. The EpiCollect entry for this date notes high flow and 240 NTU turbidity – which is not reflected in the Marstow Mill discharge data used for this plot. So, it appears that a band of rain hit the Trothy without also catching the Garren sub catchment. Clearly, using proxy discharge data has some limitations. Nevertheless, figure 6 would support an interpretation that increases in SC between 2024 and 2025 were the result of the drier summer and lower flows.

Other sampling locations show broadly similar behaviour, so further charts are not useful. Of note though (see table 3) is the fact that the middle Trothy has a slightly lower SC than do the Llymon and lower Trothy sites. This most likely reflects the slightly different geology of the upper catchment.

Of note on the Trothy sub catchment SC data is the general absence of positive spikes previously observed on SC plots in the Garren and Worm sub catchments.

Reactive Phosphorus

Again, let's take the 2025 data from FOLW020, Troy House as an exemplar. The RP trend plot in Figure 7 shows a rising trajectory through the summer months, the cause of which becomes more obvious in the RP-Q correlation plot in figure 8.

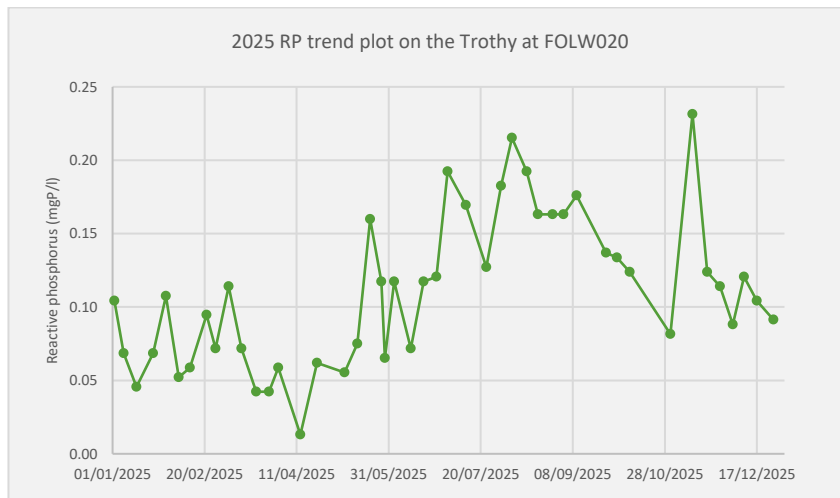


Figure 7: RP trend plot for FOLW020, Troy House, during 2025.

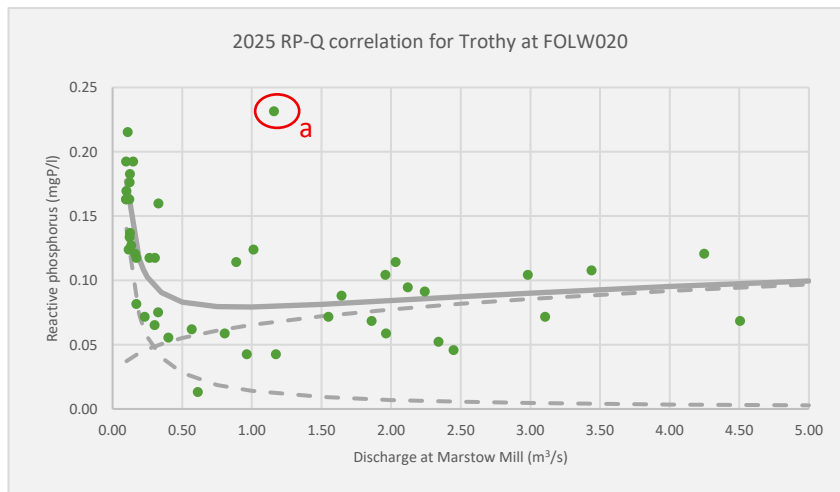


Figure 8: RP-Q correlation plot for FOLW020, Troy House, during 2025. Solid grey line is the best-fit LAM function. Dashed grey lines show the fixed-flow and diffuse components separately.

The data points are now reasonably well-behaved, and we see a very obvious rise at low flows resulting from poor dilution of fixed-flow sources. We can now apply a standard load apportionment model (LAM) represented by the following equation:

$$RP = \frac{A}{Q} + B \cdot Q^{D-1}$$

where Q is discharge, A is mass flow of RP from fixed-flow sources, B & D are parameters of the diffuse agricultural RP contribution. A, B & D are all optimised by the Excel solver to minimise residual variance. The result is the grey curve in figure 8, which can be seen to fit the data very well. The dashed lines show the two terms in the equation separately: the curve falling to the right being the first, fixed-flow term and the rising curve being the second, diffuse term. At median flow, Q_{50} (0.63 m³/s on the Garren), 28% of RP in the river is coming from sewage. During the low flows of late summer, this rose to over 70%. But at most flowrates, the Trothy is dominated by diffuse agricultural phosphorus.

Now, the fixed-flow parameter A represents a mass-flow of 1.2 kg/day of RP. Dividing that by the industry standard output per PE suggests that the sewage from circa 1000 PE is passed down the lower Trothy in 2025. Despite the acknowledged approximations introduced by the use of a proxy flow

dataset, this estimate is very consistent with our earlier estimate of likely total sewage loads. A similar analysis of the 2024 dataset yields circa 1200 PE, giving us a feel for the reproducibility of the method.

LAM analysis has been performed for each full-year RP dataset. Rather than present lots of charts, we shall summarise the results in table form.

			Measured	LAM			
Reference	Site name	Year	RP (mgP/l)	Q95 FF%	Av. Diff. (mgP/l)	Av. FF (mgP/l)	Av. FF%
Llymon Brook:							
FOLW079	Llymon Brook	2024					
		2025	0.11	65%	0.07	0.04	37%
Middle Trothy (Llanymynach to Llymon confluences)							
FOLW030	Trothy at Llantilio Crossenny	2024	0.07	93%	0.04	0.02	37%
		2025					
FOLW029	Trothy at Onen Bridge	2024	0.07	95%	0.04	0.03	41%
		2025					
Lower Trothy (below Llymon confluence)							
FOLW028	Trothy at Abbey Bridge	2024	0.10	94%	0.05	0.05	47%
		2025	0.11	66%	0.06	0.05	49%
FOLW026	Trothy at Dingestow	2024	0.08	89%	0.05	0.03	33%
		2025	0.10	80%	0.04	0.05	56%
FOLW025	Trothy at Jingle Street	2024	0.12	68%	0.09	0.02	20%
		2025	0.16	61%	0.08	0.06	44%
FOLW020	Trothy by Troy House	2024	0.10	71%	0.07	0.02	26%
		2025	0.11	71%	0.06	0.05	47%

Table 4: summary parameters from LAM analysis of RP data. Measured RP is the annual average, reproduced from table 3. Other columns are derived from the LAM analyses.

The column labelled 'Q95 FF%' indicates the LAM estimate for the contribution of fixed flow sources to the RP signal during low flow, conditions. Ranging from 61% to 95%, these are all high numbers. Sewage effluent is clearly a problem to solve in this sub catchment. Superficially, it would seem to be more of a problem in the middle Trothy than in the Llymon and lower Trothy, with figures in the region of 95%. However, closer examination shows that this is the result of a lower agricultural contribution in the middle river.

The remaining three columns show the LAM estimate for the separate contributions of fixed-flow and diffuse source RP, stated as an annual average. In other words, if we had a meter which separately measured diffuse and fixed-flow RP signals on the dates we were on the river, this is what we would have found as an annual average. Clearly, diffuse agricultural RP is the larger contribution but fixed-flow RP is highly significant. As expected, the annual average fixed-flow fraction was higher in 2025 (circa 50%) than it was in 2024 (circa 35%) due to the lower summer flows. An interesting implication of the 'Av. Diff.' column is that all sites except Jingle Street would be achieving WFD good status without the sewage contribution.

We can also make use of the LAM parameters for Troy House to estimate that during 2025, around 3000kg of RP passed down the Trothy, of which 85% was agricultural and 15% was from sewage.

Temperature

Water temperature obviously has a seasonal variation, responding to solar gain and air temperature. But maximum summer temperatures, which are the principal point of interest, tend often to be lower in smaller tributary streams than, say, in the main stem of the river Wye. This arises because of two factors. Firstly, narrow streams are more effectively shaded by any bankside vegetation present and, secondly, they are primarily groundwater fed in summer, which has a stabilising effect on river temperature.

Figure 9 shows a typical dataset, this one from the Trothy at Dingestow in 2024. But they are all very similar, with peak temperatures occurring around late July, a few weeks after the solar peak in June. It should be noted that summer temperatures have a significant diurnal cycle. Thus, inevitable inconsistencies in the timing of sampling may account for some of the scatter evident on the chart.

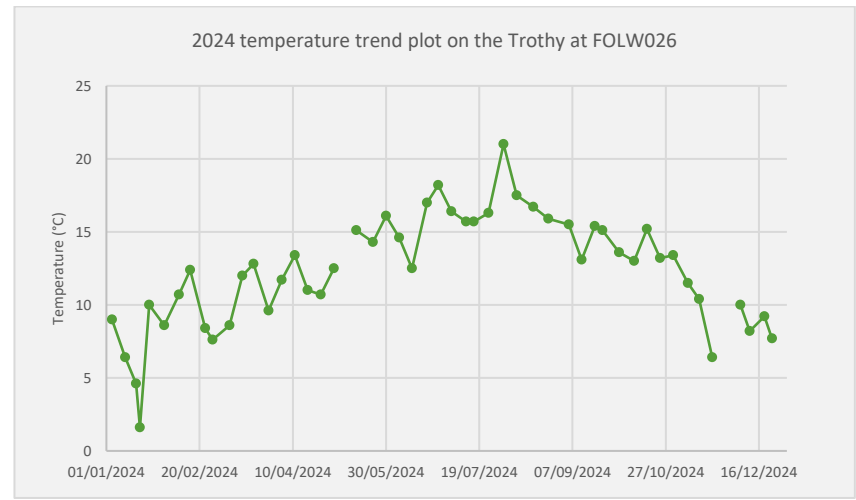


Figure 9: temperature trend plot for FOLW026, Dingestow, during 2024.

Maximum temperatures recorded at each monitoring location are shown in table 5 below. All sites monitored show very similar values. The only discernible pattern is that the overall average 2024 value of 20.5°C is a little lower than the 2025 value of 21.7°C.

Reference	Site name	Year	Maximum temperature (°C)
Llymon Brook:			
FOLW079	Llymon Brook	2024	-
		2025	22.2
Middle Trothy (Llanymynach to Llymon confluences)			
FOLW030	Trothy at Llantilio Crossenny	2024	20.7
		2025	
FOLW029	Trothy at Onen Bridge	2024	19.4
		2025	
Lower Trothy (below Llymon confluence)			
FOLW028	Trothy at Abbey Bridge	2024	20.8
		2025	
FOLW026	Trothy at Dingestow	2024	21.0
		2025	22.0
FOLW025	Trothy at Jingle Street	2024	20.4
		2025	20.1
FOLW020	Trothy by Troy House	2024	20.6
		2025	22.5

Table 4: maximum temperatures recorded at each location.

Turbidity

Turbidity data from the Trothy sub catchment up to the end of 2025 was all collected using Secchi tubes. It therefore has limitations for analytical purposes. Annual mean values tell us very little because they combine two very different flow regimes. The minimum measurable value of 12 NTU precludes analysis of the summer flow regime specifically. Also, study of the high flow regime is made difficult by our use of proxy flow rate data. Whilst Marstow Mill tracks overall trends on the Trothy reasonably well (as demonstrated by the LAM analysis of RP), the timing differences seen on high-flow events are sufficient to prevent useful turbidity-flowrate analysis of the high flow regime.

Trend plots of turbidity are generally similar from site to site. Figure 10 shows a typical plot using 2025 data from Jingle Street.

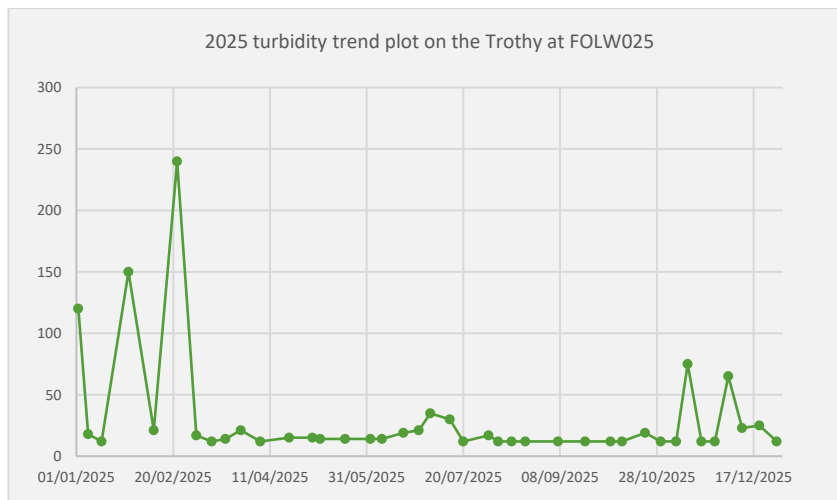


Figure 10: turbidity trend plot for FOLW025, Jingle Street, during 2025.

We find turbidity spikes in January-February and again in November-December which correspond to high-flow events (compare to the flowrate data in figure A1). Of note is the severity of these – up to 240 NTU. This suggests that soil mobilisation is not the exclusive domain of bare arable land. Heavily compacted, sloping permanent pasture can also release significant volumes of sediment, as reported by Rothamsted Research (Collins et al., 2021).

From April through to early October, the turbidity signal sticks doggedly on 12 NTU – except when it doesn't! The spike up to 35 NTU in early-July is interesting. This is very reminiscent of what we found on the Worm and almost certainly corresponds to an algal bloom. A 25 NTU spike was observed at the same time upstream at Abbey Bridge, however nothing was resolved at Dingestow or Troy House. Similar correlated spike occurred during 2024 also. In support of this, we can find observational data in EpiCollect describing green or brown coloration at different times during the summer, although entries in these fields vary from site to site and are difficult to systematically analyse.

Total Oxidised Nitrogen (TON)

TON readings within the sites reported here typically fall in the 2-5 mgN/l range, with the odd drop to 1 mgN/l. A typical TON trend plot is shown in figure 11, taken from Onen bridge in 2024. Also shown on this chart are corresponding NRW laboratory determinations (orange circles) which are in good agreement with the CS test strip data.

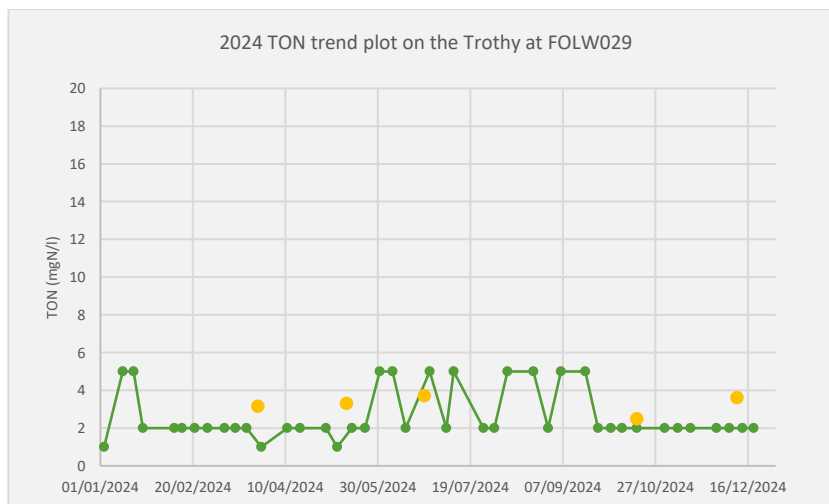


Figure 11: TON trend plot for FOLW029, Onen Bridge, during 2025. Orange circles are NRW laboratory determinations of samples from this site.

As noted earlier, this range of TON readings is noticeably lower than similar data from the Worm sub catchment (Green, 2026), which tends to be in the 5 – 10 mgN/l range. This is most probably due to the much lower fraction of arable land in the Trothy sub catchment.

There is no clear spatial variation in TON within this sub catchment, at least within the sites listed in table 1. Nor is there any obvious seasonality to this parameter, again unlike the Worm where TON is higher in the growing season.

Conclusion

- Flow modelling
 - The volumetric flow data from the Garren has proved to be a useable proxy for flowrates within the Trothy sub catchment for the purposes of LAM analysis of RP data. It is less useful for turbidity analysis due to differences in timing and magnitude of winter flow pulses.
- Reactive phosphorus
 - Most sites monitored have annual mean RP above the WFD *good* limit. Sites in the middle Trothy (Llanymynach to Llymon confluences) were slightly below the limit in 2024.
 - LAM analysis shows that, on an annual basis, most of the RP measured is diffuse agricultural pollution. The proportion of fixed-flow pollution (from STWs and PTPs) varies year to year, depending on degree of dilution in periods of low-flow. In 2024, this represented about 35% of the measured RP. But in the lower flows occurring in 2025, the fixed-flow fraction rose to around 50%. During the period of lowest flows (circa Q_{95}), the fixed-flow fraction rose well above 90% at several sites, with RP concentrations reaching 3x the WFD good threshold. Neither agriculture nor sewage loads are individually causing WFD failure on the Trothy. But, added together, the total load is too much for compliance.
 - Total annual RP load on the Trothy is around 3000 kgP, of which 85% is agricultural.
- Temperature
 - Peak temperatures appear consistent with WFD *good* status for salmonids (98%ile below 23°C).
- Turbidity
 - Secchi tube data has its limitations, and it would be beneficial to introduce the nephelometric turbidity meter at some of the Trothy sites. Nevertheless, summer algal blooms have been observed at multiple locations in the summers of both 2024 and 2025. Algal blooms at similar levels of RP and temperature have been seen in the Worm sub catchment (e.g. Jury Brook). Marginal WFD compliance is clearly not sufficient to protect from algal blooms.
- Total oxidised nitrogen
 - TON concentrations in the Trothy are around one third of the values seen on the Worm, presumably reflecting the much smaller fraction of arable farming and associated mineral fertilizer.
 - Nitrite has not been reported here, because the vast majority of measurements report zero. This is unsurprising, since NRW report nitrite in the order of 0.01mgN/l, which is below the detection limit for test strips.

Appendix A: Marstow Mill discharge data

In this report, we have used the volumetric flow (discharge) data from Marstow Mill on the lower Garren Brook. This gauging station was selected as the closest sub catchment having similar lowland topography and similar annual average rainfall. For example, the more proximate station on the Monnow was deemed to be unsuitable in view of its much flashier flows off the Black Mountains.

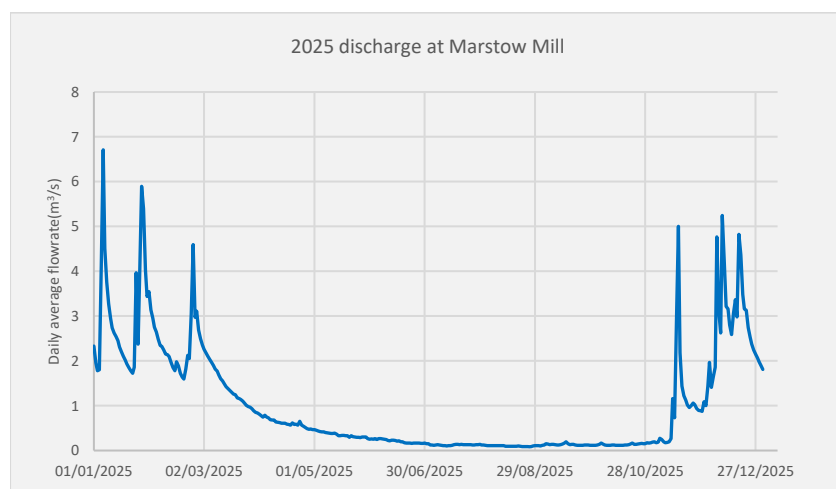


Figure A1: daily average discharge (volumetric flowrate) at Marstow Mill on the Garren Brook.

For 2 weeks spanning 13-28th August, the discharge was consistently below Q_{95} – which did not occur in 2023 or 2024. (Data downloaded from [DEFRA Hydrology Data Explorer](#)).

Of particular note in the 2025 data are the extremely low flowrates seen throughout the summer. In late August flows dipped below Q_{95} ; that is, $<0.1 \text{ m}^3/\text{s}$. By comparison, in both 2023 and 2024, the lowest flows were around $0.2 \text{ m}^3/\text{s}$. Although absolute numbers will differ, there is no reason to think that a similar pattern did not occur on the Trothy.

References

Collins, A. L., Zhang, Y., Upadhayay, H. R., Pulley, S., Granger, S. J., Harris, P., Sint, H., & Griffith, B. (2021). Current advisory interventions for grazing ruminant farming cannot close exceedance of modern background sediment loss – Assessment using an instrumented farm platform and modelled scaling out. *Environmental Science and Policy*, 116, 114–127. <https://doi.org/10.1016/j.envsci.2020.11.004>

Green, G. (2026). *Worm Brook 2025*.

Abbreviations

BFI	Base flow index: range 0-1, indicates “flashiness” of a river.
CS	Citizen science.
EA	Environment Agency.
EC	Electrical conductivity: what the HM Digital EC meters measure (but don’t display); this quantity being temperature dependent (also see SC).
LAM	Load apportionment model: an analytical model of nutrient pollution, separating the effects of fixed flow sources and diffuse agricultural sources.
NRW	Natural Resources Wales; the environmental regulator.
PTP	Private treatment plant; usually a 2-stage system rather than a simple septic tank. It is illegal for a septic tank to discharge to surface water, but many do so in rural areas.
RP	Reactive phosphorus: the phosphorus fraction reported by the Hanna meter and by laboratory determinations; this approximates to orthophosphate only, other ‘condensed’ forms not being detected.
SC	Specific conductivity: the electrical conductivity referenced to a standard temperature of 25°.
STW	Sewage treatment works.
TON	Total oxidised nitrogen, essentially nitrate plus nitrite.
WFD	Water Framework Directive; the EU legislation underpinning UK regulations for surface water quality.